

Policies and Permissions

1. Enable Multi-Factor Authentication (MFA) on your personal AWS or Azure cloud account using the built-in authenticator app option, and take a screenshot of the MFA setup confirmation page.

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Steps (AWS)

1. Log in to the **AWS Management Console**.
 2. Click your account name (top-right corner).
 3. Select **Security credentials**.
 4. Under **Multi-factor authentication (MFA)**, click **Assign MFA device**.
 5. Choose **Authenticator App**.
 6. Open an authenticator app (such as Google Authenticator or Microsoft Authenticator) on your phone.
 7. Scan the QR code displayed on the screen.
 8. Enter the two consecutive verification codes generated by the app.
 9. Click **Assign MFA**.
- **Result:** MFA is successfully enabled for your AWS account.
2. Simulate a login to your cloud account from a new browser or incognito mode and document the steps and prompts you receive with MFA enabled.

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Steps

1. Open an **Incognito/Private** browser window.
2. Go to the AWS sign-in page.
3. Enter your account ID or email address.
4. Enter your username and password.
5. After successful password verification, AWS prompts for an **MFA verification code**.
6. Open your authenticator app and enter the current 6-digit code.
7. Click **Sign In**.

❖ Observation

- With MFA enabled, logging in requires **two authentication factors**:
 - Something you know (your password)
 - Something you have (your authenticator app)
- This provides an extra layer of security and helps prevent unauthorized access.

3. Disable MFA temporarily on your cloud account, attempt to log in again, and note the difference in the login experience and security prompts.

***Hint:** Compare the number of steps and any warnings shown with and without MFA.*

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Steps

1. Log in to the AWS Management Console.
2. Go to **Security Credentials**.
3. Under **Multi-Factor Authentication (MFA)**, select your MFA device.
4. Click **Remove MFA** (or **Deactivate MFA**) and confirm the action.
5. Open another Incognito browser window.
6. Log in again using only your username and password.

With MFA Enabled

Requires username, password, and a verification code from the authenticator app.

Provides an extra layer of protection against unauthorized access.

Helps protect the account even if the password is stolen.

Without MFA

Requires only username and password.

Less secure because anyone who knows the password can sign in.

More vulnerable to phishing, password guessing, and credential theft.

4. Write a short scenario (3-5 lines) describing how an attacker might try to access a cloud account without MFA, and explain how enabling MFA would block or alert you to this attempt.

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- Suppose an attacker obtains my cloud account password through a phishing email or a leaked password database. Without MFA, the attacker can log in immediately and gain access to my cloud resources. However, if MFA is enabled, the attacker must also provide the one-time verification code generated by my authenticator app. Since only I have access to that device, the login attempt will fail. I may also receive an unexpected authentication prompt, alerting me that someone is trying to access my account, allowing me to change my password and secure the account quickly.